Amriteshkumar Yadav

Software & AI/ML Engineer

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Education

SRM Institute of Science and Technology — Chennai, India
B.Tech in Computer Science and Engineering (AI/ML Specialization), CGPA: 7.98/10 — *Graduated Jul 2026*

Technical Skills

Languages: Python, C++, C, SQL, JavaScript, TypeScript, Java, Go(Golang), Bash
Modeling: PyTorch, GNNs, XGBoost, Scikit-learn, HMM, ONNX
LLM & Vision: LLMs, RAG, LangChain, Cross-Encoders, ChromaDB, Ragas, OpenCV, YOLO
Backend, MLOps & Cloud: FastAPI, Node.js, React.js, Next.js, REST APIs, Kafka, Redis, PostgreSQL, Docker, Kubernetes, MLflow, Microservices, OpenTelemetry, Prometheus, Grafana, CI/CD, Linux, AWS, GCP

Experience

Software Engineering Intern — Jul 2025 – Jan 2026
PopFix (now SuckSi) – University-Incubated Nutraceutical Startup — *Remote*

- Spearheaded **full-stack web development** (**React.js, Next.js**) and cloud setup as the **first technical hire**, configuring production domains, DNS routing, and automated order-processing workflows from scratch.
- Developed and deployed **Python automation** scripts for order tracking, vendor coordination, and logistics workflows, reducing manual processing time by **30%** and scaling operations to **100+ monthly orders**.
- Partnered with founders in go-to-market strategy meetings as the **sole technical voice**, translating business requirements into standardized fulfillment workflows that eliminated manual spreadsheet re-keying.

Projects

Neural Database Query Optimizer (Graph Neural Networks) — GitHub
Python, PyTorch, GNNs (GCN, GAT, GraphSAGE), ONNX Runtime, PostgreSQL 16 (TPC-H), Scikit-learn, MLflow

- Trained **Graph Neural Network (GCN, GAT, GraphSAGE)** cost models in PyTorch using pairwise margin ranking loss on 184 PostgreSQL 16 `EXPLAIN` plans (12 TPC-H templates), reaching **0.74 Spearman correlation**.
- Evaluated zero-shot plan selection via **leave-one-template-out cross-validation**, outperforming gradient boosting by **+9.7 percentage points (38.9% vs. 29.2% top-1)** and achieving a **2.17x speedup** on TPC-H Q10.
- Tracked **12 architecture and pooling ablations in MLflow** and exported the winning GCN model to **ONNX Runtime**, delivering **0.21 ms p95** single-thread CPU inference (**0.44 ms p95 end-to-end** with graph parsing).

LLM Reliability Control Plane & RAG Evaluation Gateway — GitHub
Python, C++, Llama 3.2 (Ollama), LangChain, ChromaDB, Redis, FastAPI, OpenTelemetry (OTLP), DuckDB, Docker

- Built a **local RAG and hallucination guardrail pipeline** (**Llama 3.2 4-bit Q4_K_M**, ChromaDB, Pydantic) using Cross-Encoder verification, semantic similarity, and circuit-breaker fallbacks to block ungrounded outputs.
- Engineered an **OpenTelemetry (OTLP) trace ingestion gateway** in C++ and Python with CRC-32 validation and 64 MB WAL rotation, achieving **5 ms p99 ingest latency** (**<50 MB memory**) across **54 automated tests**.
- Implemented a **2-phase Zstd Parquet compaction** pipeline sustaining **50,000 spans/sec**, paired with a **Redis semantic cache** (90% cosine threshold) and **DuckDB analytical queries** completing in **<100 ms p99**.

Fault-Tolerant ML Telemetry & Feature Streaming Pipeline — GitHub
Go (Chi, Goroutines), Python, Apache Kafka, PostgreSQL 16, Redis, Prometheus, Grafana, Docker

- Engineered a concurrent **Go HTTP ingestion service** and **12-partition Apache Kafka** pipeline with batched idempotent writes, boosting throughput by **90% (288 RPS)** and cutting **p99 latency by 54.7%** vs. Python.
- Benchmarked **27 partition, consumer, and batch-size configurations** with a write-through Redis feature cache and PostgreSQL 16 storage, sustaining **7,117 events/sec (7,892 peak)** over 5+ minutes on a single host.
- Built a **Go dead-letter queue (DLQ) re-drive service** and JSON Schema validator with Prometheus metrics, routing **100% of 44,493 malformed events** with **zero data loss across 900K+ events** during container restarts.

Publications

IEEE RMKMATE 2026: *"Workplace Safety Using Computer Vision: A Framework for Personal Protective Equipment Detection and Intrusion Monitoring"* (IEEE Xplore: 11518869 | GitHub) – Co-authored a real-time YOLO and OpenCV pipeline for PPE compliance and restricted-zone intrusion tracking.

Certifications & Achievements

Oracle Generative AI Professional (2025) | Red Hat Certified System Administrator (RHCSA) | Oracle OCI Foundations Associate | Quantum Quest Expo 2024 Finalist